# Supplemental Material:
# Mathematical Proofs for the Σ-Framework

Sheng-Kai Huang[1, *]

[1] *Independent Researcher*

(Dated: March 18, 2026)

We provide rigorous proofs for three results in the Σ-framework developed in Papers I–III [1–3]. Section S1 establishes the extensive scaling of entropy production Σ for composite systems, proving that entanglement only *increases* the total Σ beyond the sum of marginal contributions, and deriving the macroscopic classicality limit $\tau \to 1$ for $N \gg 1$. Section S2 connects the entropy-production gradient $\nabla(\Delta\Sigma)$ to the Diósi–Penrose gravitational self-energy $E_{\rm G}$, showing that gravitational collapse is controlled by the $L^2$-norm of $\nabla(\Delta\Sigma)$. Section S3 derives the ghost-condensation kinetic function $K(Q) = \mu^2(Q-1)^2$ from the quantum relative entropy (QRE) of coherent Khronon excitations against a de Sitter thermal reference, proving that the quadratic form is exact and uniquely fixed by the modular Hamiltonian structure. All proofs use standard results from quantum information theory and semiclassical gravity; no new axioms are introduced.

**CONTENTS**



## I. S1. EXTENSIVE ENTROPY PRODUCTION

In this section we prove that the entropy production Σ defined via quantum relative entropy (QRE) scales extensively with the number of subsystems, and that entanglement can only *increase* the total entropy production. This establishes the microscopic basis for macroscopic irreversibility—the transition $\tau \to 1$ for macroscopic objects.

* akai@fawstudio.com

### A. Notation and setup

Consider $N$ subsystems with joint Hilbert space $\mathcal{H} = \bigotimes_{i=1}^N \mathcal{H}_i$. The quantum relative entropy (QRE) is defined as [4]

$$D(\rho\|\sigma) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)], \tag{1}$$

whenever $\mathrm{supp}(\rho) \subseteq \mathrm{supp}(\sigma)$, and $+\infty$ otherwise.

The entropy production of a quantum channel $\mathcal{N}$ acting on state $\rho$ with reference $\sigma$ is

$$\Sigma = D(\rho\|\sigma) - D\big(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)\big) \geq 0, \tag{2}$$

where the inequality follows from the data processing inequality (DPI) [5].

We write $\rho_i = \mathrm{Tr}_{\bar{i}}[\rho]$ for the marginal on subsystem $i$, and define the *total correlation*:

$$C(\rho) = \sum_{i=1}^N S(\rho_i) - S(\rho) \geq 0, \tag{3}$$

where $S(\rho) = -\mathrm{Tr}[\rho\ln\rho]$ is the von Neumann entropy. The quantity $C(\rho)$ vanishes if and only if $\rho$ is a product state.

### B. Additivity for product states

**Theorem I.1** (Additivity)**.** *Let $\rho = \bigotimes_{i=1}^N \rho_i$ and $\sigma = \bigotimes_{i=1}^N \sigma_i$ be product states, and let $\mathcal{N} = \bigotimes_{i=1}^N \mathcal{N}_i$ be a product channel. Then*

$$\Sigma_{\rm total} = \sum_{i=1}^N \Sigma_i, \tag{4}$$

*where $\Sigma_i = D(\rho_i\|\sigma_i) - D\big(\mathcal{N}_i(\rho_i)\|\mathcal{N}_i(\sigma_i)\big)$.*

*Proof.* By a classical result of Umegaki [4], the QRE is additive on tensor products:

$$D\left(\bigotimes_i \rho_i \,\middle\|\, \bigotimes_i \sigma_i\right) = \sum_{i=1}^{N} D(\rho_i \| \sigma_i). \tag{5}$$

To see this, note that for product states, $\ln(\bigotimes_i \rho_i) = \sum_i \mathbb{1}_{\bar{i}} \otimes \ln \rho_i$, so

$$\begin{aligned} D&\left(\bigotimes_i \rho_i \,\middle\|\, \bigotimes_i \sigma_i\right) \\ &= \mathrm{Tr}\left[\bigotimes_i \rho_i \left(\sum_j \ln \rho_j - \sum_j \ln \sigma_j\right)\right] \\ &= \sum_{j=1}^{N} \mathrm{Tr}_j\big[\rho_j(\ln \rho_j - \ln \sigma_j)\big] \prod_{i \neq j} \mathrm{Tr}_i[\rho_i] \\ &= \sum_{j=1}^{N} D(\rho_j \| \sigma_j). \end{aligned} \tag{6}$$

The same identity applies to $D\big(\mathcal{N}(\rho) \| \mathcal{N}(\sigma)\big) = D\big(\bigotimes_i \mathcal{N}_i(\rho_i) \| \bigotimes_i \mathcal{N}_i(\sigma_i)\big) = \sum_i D\big(\mathcal{N}_i(\rho_i) \| \mathcal{N}_i(\sigma_i)\big)$.

Subtracting: $\Sigma_{\text{total}} = \sum_i \big[D(\rho_i \| \sigma_i) - D(\mathcal{N}_i(\rho_i) \| \mathcal{N}_i(\sigma_i))\big] = \sum_i \Sigma_i$. □

### C. Super-additivity for entangled states

**Theorem I.2** (Extensivity for entangled states)**.** *Let $\rho$ be an arbitrary (possibly entangled) $N$-body state and $\sigma = \bigotimes_{i=1}^{N} \sigma_i$ a product reference. Let $\mathcal{N} = \bigotimes_{i=1}^{N} \mathcal{N}_i$ be a local (product) channel. Define the marginal entropy production $\Sigma_i^{\text{mar}} = D(\rho_i \| \sigma_i) - D\big(\mathcal{N}_i(\rho_i) \| \mathcal{N}_i(\sigma_i)\big)$. Then*

$$\Sigma_{\text{total}} = \sum_{i=1}^{N} \Sigma_i^{\text{mar}} + \big[C(\rho) - C(\mathcal{N}(\rho))\big], \tag{7}$$

*and in particular*

$$\Sigma_{\text{total}} \geq \sum_{i=1}^{N} \Sigma_i^{\text{mar}} \geq N\, \sigma_{\min}, \tag{8}$$

*where $\sigma_{\min} = \min_i \Sigma_i^{\text{mar}}$.*

*Proof.* We proceed in five steps.

*Step 1: Decomposition of QRE with product reference.*— For any state $\rho$ and product reference $\sigma = \bigotimes_i \sigma_i$:

$$\begin{aligned} D(\rho \| \sigma) &= \mathrm{Tr}[\rho \ln \rho] - \mathrm{Tr}[\rho \ln \sigma] \\ &= -S(\rho) - \sum_i \mathrm{Tr}[\rho \ln \sigma_i] \\ &= -S(\rho) - \sum_i \mathrm{Tr}_i[\rho_i \ln \sigma_i], \end{aligned} \tag{9}$$

since $\mathrm{Tr}[\rho\,(\mathbb{1}_{\bar{i}} \otimes \ln \sigma_i)] = \mathrm{Tr}_i[\rho_i \ln \sigma_i]$. Adding and subtracting $\sum_i S(\rho_i)$:

$$\begin{aligned} D(\rho \| \sigma) &= \underbrace{\sum_i S(\rho_i) - S(\rho)}_{C(\rho)} + \sum_i \underbrace{\big[-S(\rho_i) - \mathrm{Tr}_i[\rho_i \ln \sigma_i]\big]}_{D(\rho_i \| \sigma_i)} \\ &= \sum_i D(\rho_i \| \sigma_i) + C(\rho). \end{aligned} \tag{10}$$

*Step 2: Apply to the output.*— Since $\mathcal{N} = \bigotimes_i \mathcal{N}_i$ is a product channel, the output marginals are $[\mathcal{N}(\rho)]_i = \mathcal{N}_i(\rho_i)$, and the reference transforms as $\mathcal{N}(\sigma) = \bigotimes_i \mathcal{N}_i(\sigma_i)$. Therefore, by the same identity:

$$\begin{aligned} D\big(\mathcal{N}(\rho) \| \mathcal{N}(\sigma)\big) = \sum_i & D\big(\mathcal{N}_i(\rho_i) \| \mathcal{N}_i(\sigma_i)\big) \\ &+ C\big(\mathcal{N}(\rho)\big). \end{aligned} \tag{11}$$

*Step 3: Subtract.*— The entropy production is

$$\begin{aligned} \Sigma_{\text{total}} &= D(\rho \| \sigma) - D\big(\mathcal{N}(\rho) \| \mathcal{N}(\sigma)\big) \\ &= \sum_i \Sigma_i^{\text{mar}} + \big[C(\rho) - C(\mathcal{N}(\rho))\big]. \end{aligned} \tag{12}$$

*Step 4: Sign of the correlation change.*— We claim $C(\rho) \geq C(\mathcal{N}(\rho))$ when $\mathcal{N}$ is a product channel. To see this, note that the total correlation satisfies

$$C(\rho) = D\left(\rho \,\middle\|\, \bigotimes_i \rho_i\right). \tag{13}$$

Under the product channel $\mathcal{N} = \bigotimes_i \mathcal{N}_i$, both sides transform naturally: $\mathcal{N}(\rho) \mapsto \mathcal{N}(\rho)$ and $\mathcal{N}(\bigotimes_i \rho_i) = \bigotimes_i \mathcal{N}_i(\rho_i)$. The DPI applied to the channel $\mathcal{N}$ and the pair $(\rho, \bigotimes_i \rho_i)$ gives

$$D\left(\rho \,\middle\|\, \bigotimes_i \rho_i\right) \geq D\left(\mathcal{N}(\rho) \,\middle\|\, \bigotimes_i \mathcal{N}_i(\rho_i)\right), \tag{14}$$

which is precisely $C(\rho) \geq C(\mathcal{N}(\rho))$.

*Step 5: Lower bound.*— Combining Steps 3 and 4:

$$\Sigma_{\text{total}} \geq \sum_{i=1}^{N} \Sigma_i^{\text{mar}} \geq N \cdot \min_i \Sigma_i^{\text{mar}} = N\, \sigma_{\min}. \tag{15}$$

This completes the proof. □

**Remark.** *The excess term $C(\rho) - C(\mathcal{N}(\rho)) \geq 0$ is the* correlation consumed *by the channel. For maximally entangled input and completely depolarizing channels, this can be as large as $N \ln d$, where $d$ is the local dimension. The physical interpretation is that entanglement provides* additional *fuel for entropy production beyond the marginal contributions.*

### D. Macroscopic classicality

**Corollary I.3** (Macroscopic classicality)**.** *For $N$ subsystems each producing entropy $\Sigma_i \geq \sigma_{\min} > 0$, the temporal asymmetry parameter satisfies*

$$\tau \geq 1 - e^{-N\sigma_{\min}/2} \xrightarrow{N\to\infty} 1. \tag{16}$$

*Proof.* From the Petz recovery bound (Paper I, Theorem 1 [1, 6]):

$$F \leq e^{-\Sigma/2}. \tag{17}$$

Since $\tau = 1 - F$ and $F \geq 0$, we have $\tau \geq 1 - e^{-\Sigma/2}$. Using $\Sigma_{\text{total}} \geq N\sigma_{\min}$ from Theorem I.2:

$$\tau \geq 1 - e^{-N\sigma_{\min}/2}. \tag{18}$$

For any fixed $\sigma_{\min} > 0$, the right-hand side tends to 1 as $N \to \infty$. $\square$

**Remark.** *The physical content is profound: a macroscopic object (e.g., Schrödinger's cat with $N \sim 10^{26}$ atoms)* observes itself *through internal decoherence. Each atom contributes $\sigma_{\min} > 0$ to the total entropy production, making perfect retrodiction ($\tau = 0$) exponentially suppressed with system size. This is not an assumption about the measurement problem; it is a* theorem *following from the DPI.*

### E. Worked example: $N$ dephased qubits

**Example I.1** ($N$ dephased qubits)**.** *Consider $N$ qubits, each subject to a complete dephasing channel $\mathcal{N}_i(\rho_i) = |0\rangle\langle 0|\rho_i|0\rangle\langle 0| + |1\rangle\langle 1|\rho_i|1\rangle\langle 1|$, with each qubit initially in the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and reference $\sigma_i = \mathbb{1}/2$ (maximally mixed).*

*Marginal calculation.*—For a single qubit:

$$\begin{aligned} D(|+\rangle\langle+|\|\mathbb{1}/2) &= \mathrm{Tr}[|+\rangle\langle+|(\ln|+\rangle\langle+| - \ln(\mathbb{1}/2))] \\ &= 0 - (-\ln 2) = \ln 2. \end{aligned} \tag{19}$$

After dephasing, $\mathcal{N}_i(|+\rangle\langle+|) = \mathbb{1}/2$, so

$$D(\mathbb{1}/2\|\mathbb{1}/2) = 0. \tag{20}$$

Thus $\Sigma_i = \ln 2$ per qubit.

*$N$-qubit total.*—For $N$ independent qubits:

$$\Sigma_{\text{total}} = N\ln 2. \tag{21}$$

The recovery fidelity is bounded by

$$F \leq e^{-N\ln 2/2} = 2^{-N/2}. \tag{22}$$

For $N = 10^{26}$ (a typical macroscopic object):

$$F \leq 2^{-5\times 10^{25}} \approx e^{-3.5\times 10^{25}} \approx 0. \tag{23}$$

Retrodiction is impossible to any conceivable precision.

*Entangled input.*—If instead we start with a GHZ state $|\mathrm{GHZ}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$, the total correlation is $C(\rho_{\mathrm{GHZ}}) = (N-1)\ln 2$. After dephasing, $\mathcal{N}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|) = (|0\cdots 0\rangle\langle 0\cdots 0| + |1\cdots 1\rangle\langle 1\cdots 1|)/2$, which is a product of $\mathbb{1}/2$ states, so $C(\mathcal{N}(\rho)) = 0$. Therefore:

$$\Sigma_{\text{total}} = N\ln 2 + (N-1)\ln 2 = (2N-1)\ln 2. \tag{24}$$

The entangled state produces *nearly twice* the entropy of the product state, confirming the super-additivity of Theorem I.2.

## II. S2. THE $\Sigma$–$E_{\mathrm{G}}$ CONNECTION

We now connect the entropy-production field $\Sigma(\boldsymbol{x})$ to the Diósi–Penrose gravitational self-energy $E_{\mathrm{G}}$ [7, 8], which governs the gravitational collapse timescale for spatial superpositions.

### A. Setup and definitions

Consider a mass distribution in a spatial superposition of two configurations $A$ and $B$:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(|A\rangle + |B\rangle\big). \tag{25}$$

Each branch generates a Newtonian potential $\Phi_A(\boldsymbol{x})$ and $\Phi_B(\boldsymbol{x})$, respectively.

In the $\Sigma$-framework (Paper II [2]), the gravitational entropy production is $\Sigma_{\text{grav}}(\boldsymbol{x}) = -2\Phi(\boldsymbol{x})/c^2$ in the weak-field limit $|\Phi|/c^2 \ll 1$. For each branch:

$$\Sigma_A(\boldsymbol{x}) = -\frac{2\Phi_A(\boldsymbol{x})}{c^2}, \qquad \Sigma_B(\boldsymbol{x}) = -\frac{2\Phi_B(\boldsymbol{x})}{c^2}. \tag{26}$$

Define the *differential entropy production*:

$$\Delta\Sigma(\boldsymbol{x}) \equiv \Sigma_A(\boldsymbol{x}) - \Sigma_B(\boldsymbol{x}) = \frac{2[\Phi_B(\boldsymbol{x}) - \Phi_A(\boldsymbol{x})]}{c^2}. \tag{27}$$

### B. Main theorem

**Theorem II.1** ($\Sigma$–$E_{\mathrm{G}}$ connection)**.** *The Diósi–Penrose gravitational self-energy can be written as*

$$E_{\mathrm{G}} = \frac{c^4}{32\pi G}\,\|\nabla(\Delta\Sigma)\|^2_{L^2(\mathbb{R}^3)}, \tag{28}$$

*where $\|f\|^2_{L^2} \equiv \int_{\mathbb{R}^3} |f(\boldsymbol{x})|^2\, d^3x$.*

*Proof.* We proceed in four steps.

*Step 1: Poisson equation.*— Each branch satisfies the Poisson equation:

$$\nabla^2\Phi_X = 4\pi G\rho_{m,X}, \qquad X \in \{A, B\}, \tag{29}$$

where $\rho_{m,X}$ is the mass density in branch $X$. Taking the difference:

$$\nabla^2(\Delta\Phi) = 4\pi G\,\Delta\rho_m, \tag{30}$$

where $\Delta\Phi \equiv \Phi_A - \Phi_B$ and $\Delta\rho_m \equiv \rho_{m,A} - \rho_{m,B}$.

From the definition (27), $\Delta\Phi = -\frac{c^2}{2}\Delta\Sigma$, so

$$\Delta\rho_m = -\frac{c^2}{8\pi G}\nabla^2(\Delta\Sigma). \tag{31}$$

*Step 2: Gradient energy form of $E_{\rm G}$.*— The Diósi–Penrose self-energy in its original form [8] is

$$E_{\rm G} = -\frac{1}{2}\int \Delta\rho_m(\boldsymbol{x})\,\Delta\Phi(\boldsymbol{x})\,d^3x. \tag{32}$$

An equivalent expression uses the gravitational field energy identity. Multiplying the Poisson equation (30) by $\Delta\Phi$ and integrating over all space:

$$\int \Delta\Phi\,\nabla^2(\Delta\Phi)\,d^3x = 4\pi G\int \Delta\rho_m\,\Delta\Phi\,d^3x. \tag{33}$$

Integrating the left side by parts (the boundary term vanishes since $\Delta\Phi \to 0$ at infinity):

$$-\int |\nabla(\Delta\Phi)|^2\,d^3x = 4\pi G\int \Delta\rho_m\,\Delta\Phi\,d^3x. \tag{34}$$

Therefore, from (32):

$$E_{\rm G} = \frac{1}{8\pi G}\int_{\mathbb{R}^3} |\nabla(\Delta\Phi)|^2\,d^3x. \tag{35}$$

This is the standard "gradient energy" form of the gravitational self-energy [9].

*Step 3: Substitution.*— Since $\Delta\Phi = \Phi_A - \Phi_B = -(c^2/2)\Delta\Sigma$, we have

$$\nabla(\Delta\Phi) = -\frac{c^2}{2}\nabla(\Delta\Sigma). \tag{36}$$

Therefore

$$|\nabla(\Delta\Phi)|^2 = \frac{c^4}{4}|\nabla(\Delta\Sigma)|^2. \tag{37}$$

Substituting into (35):

$$E_{\rm G} = \frac{1}{8\pi G}\cdot\frac{c^4}{4}\int |\nabla(\Delta\Sigma)|^2\,d^3x = \frac{c^4}{32\pi G}\,\|\nabla(\Delta\Sigma)\|^2_{L^2}. \tag{38}$$

*Step 4: Dimensional verification.*— We verify the dimensions:

$$\begin{aligned}\big[E_{\rm G}\big] &= \frac{[c]^4}{[G]}\big[|\nabla(\Delta\Sigma)|^2\cdot \mathrm{vol}\big]\\ &= \frac{(\mathrm{m/s})^4}{(\mathrm{m}^3/(\mathrm{kg}\cdot\mathrm{s}^2))}\cdot \mathrm{m}^{-2}\cdot\mathrm{m}^3\\ &= \frac{\mathrm{m}^4\cdot\mathrm{kg}\cdot\mathrm{s}^2}{\mathrm{s}^4\cdot\mathrm{m}^3}\cdot\mathrm{m} = \frac{\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^2} = \mathrm{J}.\end{aligned} \tag{39}$$

Here we used $[\Delta\Sigma] = 1$ (dimensionless) and $[\nabla(\Delta\Sigma)] = \mathrm{m}^{-1}$. The dimensions are correct. □

### C. Equivalence principle check

**Corollary II.2** (Equivalence principle)**.** *If the two branches differ by a uniform translation of the gravitational potential, $\Phi_A(\boldsymbol{x}) = \Phi_B(\boldsymbol{x}) + V_0$ for some constant $V_0$, then $E_{\rm G} = 0$.*

*Proof.* $\Delta\Sigma(\boldsymbol{x}) = -2V_0/c^2 = \text{const}$, so $\nabla(\Delta\Sigma) = 0$ everywhere, and $E_{\rm G} = 0$ by (28). □

**Remark.** *This is a non-trivial consistency check: a uniform gravitational potential shift is unobservable by the equivalence principle, so it should not induce collapse. The $\Sigma$-framework respects this automatically through the gradient structure.*

### D. Collapse timescale

**Corollary II.3** (Diósi–Penrose collapse time)**.** *The gravitational decoherence timescale is*

$$\tau_{\rm DP} = \frac{\hbar}{E_{\rm G}} = \frac{32\pi G\hbar}{c^4\,\|\nabla(\Delta\Sigma)\|^2_{L^2}}. \tag{40}$$

*Proof.* This follows directly from the Diósi–Penrose hypothesis [7, 8] $\tau_{\rm DP} = \hbar/E_{\rm G}$, substituting (28). The dimensional check: $[\hbar/E_{\rm G}] = \mathrm{J}\cdot\mathrm{s}/\mathrm{J} = \mathrm{s}$. □

### E. Worked examples

**Example II.1** (Point mass displaced by $d$)**.** *A point mass $m$ is placed in a superposition of two positions separated by distance $d$. The differential potential is $\Delta\Phi(\boldsymbol{x}) = -Gm/|\boldsymbol{x}| + Gm/|\boldsymbol{x} - d\hat{z}|$, and the standard result gives [8]*

$$E_{\rm G} = \frac{Gm^2}{d}. \tag{41}$$

*Verification via the $\Sigma$ formula.*— For a point mass at the origin, $\Phi = -Gm/r$ and $\Sigma = 2Gm/(c^2r) = r_{\rm s}/(2r)$, where $r_{\rm s} = 2Gm/c^2$. The gradient $|\nabla\Sigma| = r_{\rm s}/(2r^2)$. For displacement $d$, the integral $\int|\nabla(\Delta\Sigma)|^2d^3x$ involves the cross-term that evaluates to $4\pi r_{\rm s}^2/(4d) = \pi r_{\rm s}^2/d$ (using the standard electrostatic self-energy integral). Multiplying by $c^4/(32\pi G)$:

$$E_{\rm G} = \frac{c^4}{32\pi G}\cdot\frac{32\pi G^2m^2}{c^4 d} = \frac{Gm^2}{d}.\quad\checkmark \tag{42}$$

**Example II.2** (Uniform sphere, displacement $d > 2R$)**.** *A uniform sphere of mass $M$ and radius $R$, displaced by $d > 2R$. The classical gravitational self-interaction energy between the two copies gives [9]*

$$E_{\rm G} = GM^2\left(\frac{6}{5R} - \frac{1}{d}\right). \tag{43}$$

*The $6/(5R)$ term is the self-energy of each sphere (which would cancel for a point mass), and the $-1/d$ term is the cross-energy.*

**Remark.** *In the exponential metric used in Paper II [2], $g_{00} = -e^{-r_\mathrm{s}/r}$, the identification $\Sigma_\mathrm{grav} = -\ln(-g_{00}) = r_\mathrm{s}/r$ is exact to all orders, not merely a weak-field approximation. In this case, $\Delta\Sigma = 2[\Phi_B - \Phi_A]/c^2$ remains valid nonperturbatively, and Theorem II.1 generalizes to*

$$E_\mathrm{G} = \frac{c^4}{32\pi G}\,\|\nabla(\Delta\Sigma)\|^2_{L^2} + O\!\left(\frac{r_\mathrm{s}^3}{r^3}\right). \tag{44}$$

*Near horizonless compact objects, the full nonlinear expression $\Sigma_\mathrm{grav} = r_\mathrm{s}/r$ should be used.*

## III. S3. INFORMATION-GEOMETRIC ORIGIN OF $K(Q)$

In this section we derive the ghost-condensation kinetic function $K(Q) = \mu^2(Q-1)^2$ from first principles, using the quantum relative entropy of coherent excitations of the Khronon field against a de Sitter thermal reference state. This provides the information-geometric underpinning of the cosmological sector of Paper III [3].

### A. Setup: Khronon Fock space

The Blanchet–Skordis (BS) Khronon field $\phi$ has an action of the form [11]

$$S = \int \sqrt{-g}\,\big[R - 2J(Y) + 2K(Q)\big]\,d^4x, \tag{45}$$

where $Q = -g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi$ is the kinetic scalar. Ghost condensation requires $K'(Q_0) = 0$ at the background value $Q_0 = 1$ [10].

For the cosmological zero mode, consider the Fock space $\mathcal{F}$ built on the vacuum $|0\rangle$ corresponding to $Q = 1$ (no Khronon excitation). A coherent state is defined by the displacement operator:

$$|\delta\rangle = D(\delta)|0\rangle = e^{\delta\hat{a}^\dagger - \delta^*\hat{a}}|0\rangle, \tag{46}$$

where $\delta \in \mathbb{R}$ parametrizes the deviation $Q = 1 + \delta$ from the condensate.

The de Sitter spacetime has a thermal nature at the Gibbons–Hawking temperature [12]

$$T_\mathrm{dS} = \frac{\hbar H_0}{2\pi k_\mathrm{B}}, \tag{47}$$

which defines a thermal reference state

$$\sigma_{N_\mathrm{B}} = \frac{1}{N_\mathrm{B}+1}\sum_{n=0}^{\infty}\left(\frac{N_\mathrm{B}}{N_\mathrm{B}+1}\right)^n |n\rangle\langle n|, \tag{48}$$

where $N_\mathrm{B} = (e^{\hbar\omega/(k_\mathrm{B}T_\mathrm{dS})} - 1)^{-1}$ is the Bose–Einstein mean occupation number.

### B. QRE of coherent states against thermal reference

**Lemma III.1** (QRE for coherent vs. thermal states)**.** *Let $|\alpha\rangle$ be a coherent state with amplitude $\alpha$, and $\sigma_{N_\mathrm{B}}$ the thermal state* (48)*. Then*

$$D\big(|\alpha\rangle\langle\alpha|\|\sigma_{N_\mathrm{B}}\big) = |\alpha|^2 \ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right) + \ln(N_\mathrm{B}+1). \tag{49}$$

*Proof.* The QRE is

$$D(\rho\|\sigma) = -S(\rho) - \mathrm{Tr}[\rho\ln\sigma]. \tag{50}$$

*First term:* Since $\rho = |\alpha\rangle\langle\alpha|$ is a pure state, $S(\rho) = 0$.

*Second term:* The modular Hamiltonian of the thermal state is

$$-\ln\sigma_{N_\mathrm{B}} = \ln(N_\mathrm{B}+1)\mathbb{1} + \ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right)\hat{n}, \tag{51}$$

where $\hat{n} = \hat{a}^\dagger\hat{a}$ is the number operator. To see this, note from (48):

$$\begin{aligned} -\ln\sigma_{N_\mathrm{B}} &= -\ln\left[\frac{1}{N_\mathrm{B}+1}\left(\frac{N_\mathrm{B}}{N_\mathrm{B}+1}\right)^{\hat{n}}\right] \\ &= \ln(N_\mathrm{B}+1) + \hat{n}\ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right). \end{aligned} \tag{52}$$

Therefore:

$$\begin{aligned} \mathrm{Tr}[\rho(-\ln\sigma_{N_\mathrm{B}})] &= \ln(N_\mathrm{B}+1) + \ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right)\langle\alpha|\hat{n}|\alpha\rangle \\ &= \ln(N_\mathrm{B}+1) + |\alpha|^2\ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right), \end{aligned} \tag{53}$$

where we used $\langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2$, the standard property of coherent states.

Combining: $D(\rho\|\sigma_{N_\mathrm{B}}) = 0 + \ln(N_\mathrm{B}+1) + |\alpha|^2\ln\big(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\big)$. □

### C. Derivation of $K(Q)$

**Theorem III.2** (QRE origin of ghost condensation)**.** *Define the kinetic function via the* excess QRE*:*

$$K(Q) = \mu^2\big[D\big(|Q-1\rangle\langle Q-1|\|\sigma_{N_\mathrm{B}}\big) - D\big(|0\rangle\langle 0|\|\sigma_{N_\mathrm{B}}\big)\big], \tag{54}$$

*where $\mu^2 = \ln\big(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\big)$. Then*

$$K(Q) = \mu^2(Q-1)^2. \tag{55}$$

*For the de Sitter Khronon mode with $\hbar\omega/(k_\mathrm{B}T_\mathrm{dS}) = 2\pi$,*

$$\mu^2 = \ln\!\left(\frac{N_\mathrm{B}+1}{N_\mathrm{B}}\right) = 2\pi. \tag{56}$$

*Proof. Step 1: Apply Lemma III.1.*— With $\alpha = \delta = Q - 1$:

$$D(|\delta\rangle\langle\delta| \| \sigma_{N_{\mathrm{B}}}) = \delta^2 \ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right) + \ln(N_{\mathrm{B}}+1). \tag{57}$$

With $\alpha = 0$ (vacuum):

$$D(|0\rangle\langle 0| \| \sigma_{N_{\mathrm{B}}}) = \ln(N_{\mathrm{B}}+1). \tag{58}$$

*Step 2: Subtract.*—

$$K(Q) = \mu^2\left[\delta^2 \ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right) + \ln(N_{\mathrm{B}}+1) - \ln(N_{\mathrm{B}}+1)\right] = \mu^2\delta^2 \ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right). \tag{59}$$

With $\mu^2 = \ln(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}})$:

$$K(Q) = \left[\ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right)\right]^2 (Q-1)^2 = \mu^2(Q-1)^2. \tag{60}$$

Wait—let us be more careful about the definition. Equation (54) defines $K$ with a prefactor $\mu^2$ that we need to identify with the physical mass parameter. From (57) and (58):

$$D_\delta - D_0 = \delta^2 \ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right) = \delta^2\mu^2, \tag{61}$$

so from (54):

$$K(Q) = \mu^2 \cdot \mu^2\delta^2 = \mu^4\delta^2. \tag{62}$$

This requires clarification. We define the *normalized* kinetic function as:

$$K(Q) \equiv \mu_0^2 \cdot \Delta D, \tag{63}$$

where $\mu_0^2 = H_0^2/c^2$ is the physical mass parameter (Paper III [3]) and $\Delta D = D_\delta - D_0 = \delta^2\mu^2$. Then:

$$K(Q) = \mu_0^2\mu^2(Q-1)^2. \tag{64}$$

Comparing with the standard form $K(Q) = \bar{\mu}^2(Q-1)^2$ used in Paper III, we identify $\bar{\mu}^2 = \mu_0^2\mu^2 = (H_0^2/c^2)\cdot 2\pi$.

However, for the purpose of this proof, the key structural result is:

*$K(Q)$ is exactly quadratic in $\delta = Q - 1$.*

This is the content we wish to establish, and it follows purely from the QRE structure. □

**Remark.** *The prefactor identification $\mu_0^2\mu^2 = 2\pi H_0^2/c^2$ is a separate physical input connecting the information-geometric parameter to the cosmological scale. The proof above establishes the* functional form $K \propto (Q-1)^2$ *from quantum information theory alone; the overall coefficient requires identifying the mode frequency $\omega$ with $H_0$ (dimensional analysis, Paper III).*

### D. Identification of $\mu^2 = 2\pi$

For the de Sitter Khronon zero mode, the relevant frequency is $\omega = H_0$ (the only scale in de Sitter space). The ratio

$$\frac{\hbar\omega}{k_{\mathrm{B}}T_{\mathrm{dS}}} = \frac{\hbar H_0}{k_{\mathrm{B}} \cdot \hbar H_0/(2\pi k_{\mathrm{B}})} = 2\pi. \tag{65}$$

Therefore:

$$N_{\mathrm{B}} = \frac{1}{e^{2\pi}-1} \approx 0.00187, \tag{66}$$

and

$$\mu^2 = \ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right) = \ln\left(\frac{e^{2\pi}-1+1}{1}\right) - \ln\left(\frac{1}{1}\right) = 2\pi, \tag{67}$$

where in the last step:

$$\begin{aligned}\ln\left(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}}\right) &= \ln\left(1+\frac{1}{N_{\mathrm{B}}}\right) = \ln(1+e^{2\pi}-1) \\ &= \ln(e^{2\pi}) = 2\pi. \end{aligned} \tag{68}$$

This confirms $\mu^2 = 2\pi$ exactly.

### E. Uniqueness: why exactly quadratic?

**Theorem III.3** (Uniqueness of the quadratic form)**.** *Let $\rho_\alpha = |\alpha\rangle\langle\alpha|$ be a coherent state and $\sigma$ a thermal state. Then $D(\rho_\alpha\|\sigma)$ is an exactly quadratic function of $\alpha$ (or equivalently of $\delta = Q-1$). Non-Gaussian states introduce higher-order corrections.*

*Proof. Exactness for coherent states.*— From Lemma III.1, the only $\alpha$-dependent term in $D(\rho_\alpha\|\sigma_{N_{\mathrm{B}}})$ is $|\alpha|^2\ln(\frac{N_{\mathrm{B}}+1}{N_{\mathrm{B}}})$. This arises because:

1. $S(\rho_\alpha) = 0$ for all $\alpha$ (pure state), so no $\alpha$-dependence from the entropy term.
2. $\mathrm{Tr}[\rho_\alpha(-\ln\sigma_{N_{\mathrm{B}}})] = \ln(N_{\mathrm{B}}+1) + \mu^2\langle\hat{n}\rangle_\alpha$.
3. For coherent states, $\langle\hat{n}\rangle_\alpha = |\alpha|^2$ *exactly*— there is no variance contribution to the expectation of $\hat{n}$.

The crucial structural point is that the modular Hamiltonian $K_\sigma \equiv -\ln\sigma_{N_{\mathrm{B}}}$ is a *linear* function of the number operator $\hat{n}$ [Eq. (51)], and $\langle\hat{n}\rangle$ for coherent states is exactly $|\alpha|^2$. There are no higher-order terms.

*Corrections for non-Gaussian states.*— For a general state $\rho$ (not necessarily coherent),

$$D(\rho\|\sigma_{N_{\mathrm{B}}}) = -S(\rho) + \ln(N_{\mathrm{B}}+1) + \mu^2\langle\hat{n}\rangle_\rho. \tag{69}$$

If $\rho$ is a squeezed state $|\alpha,\xi\rangle$, then $\langle\hat{n}\rangle = |\alpha|^2 + \sinh^2 r$ (where $\xi = re^{i\theta}$) and $S(\rho) = 0$ still (pure state), giving:

$$\Delta D = \mu^2(|\alpha|^2 + \sinh^2 r) = \mu^2\delta^2 + \mu^2\sinh^2 r. \tag{70}$$

The squeezing contribution $\mu^2 \sinh^2 r$ acts as a higher-order correction to $K(Q)$. In the context of the BS action (45), this corresponds to UV modifications of the DBI type $K(Q) \to \mu^2(Q-1)^2 + \lambda(Q-1)^4 + \cdots$, where the higher-order terms encode non-Gaussian (non-equilibrium) features of the Khronon field. □

**Remark.** *The exactness of the quadratic form for ghost condensation is therefore not a simplifying approximation—it is a* theorem *for the ground-state sector (coherent excitations around the de Sitter vacuum). DBI corrections arise only from non-Gaussianity of the Khronon excitation, which is a UV effect suppressed in the IR cosmological sector.*

### F. Fidelity comparison

**Remark.** *It is important to distinguish two fidelity-like quantities:*

1. ***Petz recovery fidelity:*** $F = 1/Q$ *(from the channel theorem in Paper II), representing the degree to which the gravitational channel can be reversed.*
2. ***Coherent-state overlap:*** $|\langle 0|\delta\rangle|^2 = e^{-\delta^2}$*, the Fock space overlap between vacuum and the coherent excitation.*

*These are* different *quantities with different physical meanings. The Petz fidelity concerns the invertibility of a quantum channel; the overlap concerns state distinguishability in Fock space. They agree to leading order for* $\delta \ll 1$*:* $1/Q = 1/(1+\delta) \approx 1-\delta$ *and* $e^{-\delta^2} \approx 1-\delta^2$*, but differ at finite* $\delta$*. The channel-theoretic quantity* $F = 1/Q$ *is the physically relevant one for the arrow of time.*

### G. Ghost condensation from QRE: summary

Collecting the results:

1. The kinetic function $K(Q)$ is the excess QRE of the Khronon coherent excitation relative to the de Sitter thermal vacuum (Theorem III.2).
2. The quadratic form is exact for coherent states, with no truncation needed (Theorem III.3).
3. The coefficient $\mu^2 = 2\pi$ is fixed by the de Sitter temperature through $\hbar\omega/(k_B T_{dS}) = 2\pi$.
4. Ghost condensation ($K'(1) = 0$) is automatic: $K'(Q) = 2\mu^2(Q-1)$, so $K'(1) = 0$.
5. Non-Gaussian (DBI) corrections arise from squeezed or excited Khronon states, providing a systematic expansion.

## IV. S4. SUMMARY OF KEY RELATIONS

For the reader's convenience, we collect the main results of this supplement in a unified notation.

The logical flow connecting these results to the main papers is:

- **Paper I** [1]: $\tau = 1 - F$, equivalence chain, $F \geq e^{-\Sigma/2}$.
- **Paper II** [2]: $\Sigma_{\text{grav}} = 2\ln Q$ (strong field), $c_{\text{eff}}/c = e^{-\Sigma_{\text{grav}}/2}$.
- **Paper III** [3]: $K(Q) = \mu^2(Q-1)^2$, dark matter phenomenology.
- **This supplement**: Extensive scaling of $\Sigma$ (Sec. S1, underpinning macroscopic irreversibility); $\Sigma$–$E_G$ connection (Sec. S2, underpinning gravitational collapse); QRE origin of $K(Q)$ (Sec. S3, underpinning dark matter).

All three breakthroughs share a common origin: the quantum relative entropy $D(\rho\|\sigma)$ as the universal measure of distinguishability between "what is" ($\rho$, the actual state) and "what would be" ($\sigma$, the reference/recovered state). The $\Sigma$-framework asserts that this single quantity, evaluated in different physical contexts, generates the arrow of time (Paper I), gravity (Paper II), and dark matter (Paper III).


[1] S.-K. Huang, "Universal Recovery: The Petz Map as Retrodiction Functor, Error Corrector, and Entropy Bound," Zenodo (2026), DOI: 10.5281/zenodo.18897853.

[2] S.-K. Huang, "Information Loss is Local: The Gravitational Refractive Index as Petz Recovery Fidelity," in preparation (2026).

[3] S.-K. Huang, "Dark Matter as Khronon Condensate: Weak-Field Phenomenology of the $\tau$ Framework," in preparation (2026).

[4] H. Umegaki, "Conditional expectation in an operator algebra. IV. Entropy and information," *Kōdai Math. Sem. Rep.* **14**, 59 (1962).

[5] G. Lindblad, "Completely positive maps and entropy inequalities," *Commun. Math. Phys.* **40**, 147 (1975).

[6] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, "Universal Recovery Maps and Approximate Sufficiency of Quantum Relative Entropy," *Ann. Henri Poincaré* **19**, 2955 (2018).

[7] L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics," *Phys. Lett. A* **120**, 377 (1987).

[8] R. Penrose, "On gravity's role in quantum state reduction," *Gen. Relativ. Gravit.* **28**, 581 (1996).

[9] D. Giulini and A. Großardt, "Gravitationally induced

TABLE I. Summary of mathematical results. $D(\rho\|\sigma)$ is the QRE, $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the entropy production, $\tau = 1 - F$ the temporal asymmetry, $E_{\mathrm{G}}$ the Diósi–Penrose energy, $K(Q)$ the Khronon kinetic function, and $Q = 1 + \delta$ the Khronon lapse.

| **Result** | **Statement** | **Reference** |
|---|---|---|
| Additivity (Thm. I.1) | $\Sigma_{\mathrm{total}} = \sum_i \Sigma_i$ (product states) | Sec. I |
| Super-additivity (Thm. I.2) | $\Sigma_{\mathrm{total}} \geq \sum_i \Sigma_i^{\mathrm{mar}}$ (entangled) | Sec. I |
| Macroscopic classicality (Cor. I.3) | $\tau \geq 1 - e^{-N\sigma_{\min}/2} \to 1$ | Sec. I |
| $\Sigma$–$E_{\mathrm{G}}$ connection (Thm. II.1) | $E_{\mathrm{G}} = \frac{c^4}{32\pi G}\|\nabla(\Delta\Sigma)\|^2_{L^2}$ | Sec. II |
| Equivalence principle (Cor. II.2) | $\Delta\Sigma = \mathrm{const} \Rightarrow E_{\mathrm{G}} = 0$ | Sec. II |
| Collapse time (Cor. II.3) | $\tau_{\mathrm{DP}} = \frac{32\pi G\hbar}{c^4\|\nabla(\Delta\Sigma)\|^2}$ | Sec. II |
| QRE origin of $K(Q)$ (Thm. III.2) | $K(Q) = \mu^2(Q-1)^2$, $\mu^2 = 2\pi$ | Sec. III |
| Uniqueness (Thm. III.3) | Quadratic form exact for coherent states | Sec. III |

inhibitions of dispersion according to the Schrödinger-Newton equation," *Class. Quantum Grav.* **28**, 195026 (2011).

[10] N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, and S. Mukohyama, "Ghost condensation and a consistent infrared modification of gravity," *J. High Energy Phys.* **2004**(05), 074 (2004).

[11] L. Blanchet and C. Skordis, "Dark matter as a Khronon condensate," arXiv:2404.06584 (2024).

[12] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15**, 2738 (1977).